Canada will remain a leading spacefaring nation, establishes the Canadian Space Agency in its Space Strategy for Canada, by participating in the next effort to push humanity into space: the Lunar Gateway Mission [4]. Building on the success of the Canadarms, Canada's participation will materialize as a next-generation artificial intelligence-enabled deep-space robotic system for the Lunar Gateway Mission, a milestone that will allow Canada to strengthen its position as a world leader in AI-enabled space robotics and position Canadian firms as global leaders in the future on-orbit servicing market [4].

On-orbit servicing, in particular, is embedded within one ambitious future concept of operations for the Lunar Gateway where it interacts with other spacecraft and their payloads for applications including refueling, inspection, assembly, and maintenance [7, 2]. In such scenarios, servicing is generally thought to be preceded by rendezvous and proximity operations, and may be both preceeded and succeeded by a capture/release phase like docking, grappling or berthing [1]. Proximity operations is a particularly challenging phase that in turn comprises the awareness and navigation capabilties leveraged when two spacecraft operate close to each other [3, 6, 5, 8].

The research community argues that awareness in proximity operations, the capability supported by this project, should be adaptable to targets ranging from cooperative spacecraft to non-cooperative debris. This must be accomplished without compromising robustness to the tight energy constraints of spaceborne systems and the ample variations in space conditions such as lighting and electromagnetic noise [8, 8, 6, 5, 9].

Cameras, particularly the common monocular, red-green-blue light intesity sensing type, have been put forward as the sensor of choice for proximity operations awareness owing to their relative energy efficiency, high frame rate, and low cost [6, 5]. Space institutions have consequently challenged researchers to exploit images as the input for awareness problems like pose estimation and object detection, as evidenced by the 2019 and 2021 editions of the Satellite Pose Estimation Competition hosted by NASA and ESA, and NASA's 2024 Pose Bowl Spacecraft Detection and Pose Estimation Challenge. The academic community has

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